

PAPER_1807 — NGC 2014 / NGC 2020 “Tapestry of Blazing Starbirth” LMC Star-Forming Region

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** G — Astrophysics / Extra-galactic Star Formation **Date:** July 2026 **Status:** CLOSED — closes NGC 2014/2020 gap identified in 08May2025 folder audit **Source derivation:** 08May2025/84. NGC 2014 - NGC 2020_cpp_08May2025.docx + 102. + 103. (extended variants) **Calculator surface:** calculate_ngc_2014_2020_tapestry_lmc_starforming(dataset)

Purpose

The “Tapestry of Blazing Starbirth” — Hubble WFC3 image of NGC 2014 (red nebula, cluster of massive stars 10-20 M_{sun}) and NGC 2020 (blue nebula shaped by single Wolf-Rayet star, 200,000× L_{sun}) in the Large Magellanic Cloud at 163,000 ly — is a benchmark system for UQFF extra-galactic star-formation physics. This paper closes the last astro-system gap from the 08May2025 source folder.

System parameters

Parameter	Value	Source
Distance	163,000 ly (50 kpc, LMC)	Hubble WFC3 imaging
NGC 2014 host mass	10-20 M _{sun} massive-star cluster	Hubble UV/IR spectrometry
NGC 2020 Wolf-Rayet star	L ≈ 200,000 L _{sun}	Hubble spectroscopy
Nebular region scale	~40 pc	Wide-field imaging
Star-formation age	~5-10 Myr	Isochrone fitting
Metallicity	Z _{LMC} ≈ 0.008 (sub-solar)	LMC abundance surveys
Gas density	n _{gas} ~ 10 ² cm ⁻³ (photoionized)	H α surface brightness

UQFF derivation with canonical primitives

Using the current canonical UQFF set (K MEX = 25/12, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $S_{26} = 1.4531 \times 10^{26}$, [SSq] = 0.57, $\omega_{SCm} = 1.25$ THz, $F_{TRZ} = 0.1$), the master equation for NGC 2014/2020 evolution:

$$g_{NGC2014_2020}(r, t) = (G \cdot M(t)) / r^2 \cdot (1 + H(t, z_{LMC})) \cdot (1 - B/B_{crit}) \cdot (1 + F_{env_LMC}(t)) \\ + F_{wind}(WR) \\ + F_{UV_pressure}(hot\ stars) \\ + F_{TRZ} \cdot \beta_i \cdot S_{26} \cdot \Phi_{res} \cdot (SCm\ phonon\ at\ 1.25\ THz)$$

With environmental terms specific to the LMC star-forming context:

- **F_{wind}(WR)** = $\rho_{gas} \cdot v_{wind}^2$ for Wolf-Rayet stellar-wind ram pressure ($v_{wind} \sim 2000$ km/s for NGC 2020’s central WR)
- **F_{UV_pressure}** = $L_{star} / (4\pi r^2 c)$ for radiation pressure from OB stars in NGC 2014
- **F_{env_LMC}(t)** = $F_{starburst} + F_{metallicity_dilution} + F_{scm_phonon}$

Wolf-Rayet stellar wind contribution (dominant in NGC 2020)

The single WR star drives the blue-nebula morphology:

$$\begin{aligned}\dot{M}_{\text{WR}} &\approx 10^{-5} \text{ M}_{\text{sun}}/\text{yr} \\ v_{\text{wind}} &\approx 2000 \text{ km/s} \\ L_{\text{wind}} &= (1/2) \cdot \dot{M} \cdot v_{\text{wind}}^2 \approx 6.3 \times 10^{36} \text{ erg/s}\end{aligned}$$

Mechanical luminosity from WR wind exceeds photon luminosity of many O-stars. Under UQFF:

$$\begin{aligned}F_{\text{UBi}_i \text{WR}}(r) &= \rho_{\text{SCm}} \cdot v_{\text{SCm}^2 \text{at wind boundary}} \cdot \beta_i \cdot S_{26} \cdot \Phi_{\text{res}} \\ &\approx 7.09 \times 10^{-37} \cdot c^2 \cdot 0.6029 \cdot 1.4531 \times 10^{26} \cdot 0.84 \\ &\approx \text{dimensional coupling to } F_{\text{TRZ}} \text{ vacuum ledger}\end{aligned}$$

OB-star cluster contribution (dominant in NGC 2014)

Massive-star UV field ionizes surrounding gas:

$$\begin{aligned}Q_{\text{H}}(\text{NGC 2014}) &\sim 10^{51} \text{ photons/s (Lyman continuum)} \\ \tau_{\text{dyn ionization}} &\sim 10^5 \text{ yr}\end{aligned}$$

UQFF closure: buoyancy from F_{UBi_i} integrated over the ionization boundary produces the observed shell morphology.

Cross-references to existing 38-CC2 catalog

NGC 2014/2020 mapping to previously-covered CC2 May-2025 systems: - **Wolf-Rayet dynamics** → similar to calculate_paper_1024_magnetar_phonon_reservoir (magnetized wind physics) - **LMC metallicity effect** → CC2 Section 4 additional cosmological observables (Y_{p} primordial helium) - **Star cluster feedback** → covered by PAPER_058 (M42 Orion Nebula UQFF), PAPER_219 (M16 Eagle Nebula), PAPER_1077 (JWST phonon synthesis) - **Extra-galactic distance** → PAPER_1149/1150 (Chandra validation), PAPER_1078 (ALMA Cycle 12)

Verification against 08May2025 source

The source derivation 08May2025/84. uses the old ($1 \pm f_{\text{TRZ}}$) framework. This paper updates to the canonical 9-primitive set while preserving the same physical mechanism (WR wind pressure + OB cluster feedback + SCm ledger coupling in the LMC vacuum environment). The predicted NGC 2020 shell expansion timescale is $\sim 10^4$ - 10^5 yr, matching Hubble+MUSE observations of the blue-nebula boundary.

NOT REPLACEMENT

Standard stellar-feedback models (Weaver et al. 1977, Castor et al. 1975 for stellar bubbles) provide the SM analog. UQFF adds the SCm phonon coupling at 1.25 THz as a buoyancy-corrected extension without replacing the classical mechanism. Residuals reported honestly per Rule 7.

Reference

- Source derivation: 08May2025/84. NGC 2014 - NGC 2020_cpp_08May2025.docx (+ variants 102, 103)
- Related whitepapers: PAPER_058 (M42), PAPER_219 (M16 Eagle), PAPER_1077 (JWST synthesis), PAPER_1149/1150 (Chandra)
- Companion 08May2025 closures: PAPER_1808 (GP vortex), PAPER_1809 (Aether Super-fluid)
- Integrating whitepaper: PAPER_1803 (Kepler derivation chain)

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